Vario USB — BMP581 + LSM6DSV16X

Consolidated Variometer Engineering Validation Report

Platform	ESP32-S3 @ 240 MHz, ESP-IDF firmware "vario-idf 0.2.0"
Barometer	Bosch BMP581, I2C bus B (GPIO6/7) @ 0x47, ODR 40 Hz, IIR bypass
IMU	ST LSM6DSV16X, I2C bus A (GPIO4/5) @ 0x6B, 240 Hz, +-8 g / +-500 dps
Estimators (frozen configs)	D01 (CV Kalman), C03 (CV Tracker), S4 (Fast/Slow), S5 (Light Gate), FUS (baro+IMU fusion)
Virtual audio policy	lift_on +0.20 / lift_off +0.05 m/s, tone 500 + 450*v Hz
Record under review	20261025_225610_idf_bmp581LSM6DS_elevator_2x_200s (180.1 s device span, four legs 14->1->14->1->14)
Report date	2026-10-25
Status	Engineering study — no estimator selected or frozen

All quantitative statements are either directly measured from recorded telemetry, pressure-derived reference estimates, or reproduced from the cited frozen-configuration benchmark phases. Pressure-derived timings are relative comparisons, not absolute ground-truth latency measurements.

1 Executive summary and key findings

This report consolidates the complete measurement and benchmark record of the Vario USB programme through the first fully-validated BMP581 + LSM6DSV16X acquisition: sensor bring-up, a root-caused I2C failure mode, three static tests, a hand test, and a clean four-leg elevator recording (22:56:10, 2026-10-25), together with the historical filter-architecture benchmarks (Phases 2-7) against which the new datasets are compared.

- • **Acquisition integrity is established.** The principal elevator recording is flawless: zero sequence drops, zero malformed lines, zero I2C errors, IMU 100.006 Hz (max gap 11.9 ms, zero stale frames), baro 40.006 Hz, host/device clock agreement within 0.2 s.
- • **DEF-020 was root-caused and fixed.** The mid-session IMU death (~25 s) traced to bus A running at 400 kHz with only the ESP32-S3 weak (~45 kOhm) internal pull-ups. Lowering bus A to 100 kHz (BMP581 bus untouched) gives zero IMU errors across 60 s + 90 s verification, 100.0 Hz streaming and 0.0 % stale samples. The collateral BMP581 slowdown (40 -> 31.25 Hz) was task-time contention, not a bus fault.
- • **Baro-only estimators are effectively tied on elevator dynamics.** D01/C03 lead audio alert timing (ON ~0.35-0.45 s after the pressure-derived motion threshold); S4/S5 lag by 0.1-0.35 s; tone-derived plateau velocities match the pressure reference to within ~0.01 m/s mean (MAE 0.06-0.07 m/s). Zero false audio episodes on descents and floor dwells.
- • **FUS (IMU fusion) is measurably worse on this data.** It starts ~0.75-1.2 s late, overshoots sustained climb by ~+0.20 m/s, and stays ON ~5-6 s after each top-floor stop. Consistent with the Bench 7 finding: quiet-building data has no vibration for IMU-aiding to cancel. This is an estimator/configuration observation, not an acquisition defect.
- • **Static audio cleanliness is unprecedented.** No candidate produced a single false audio ON in the 90 s BMP581 static (Bench 7, 37 frozen configurations); the same holds for the new-era statics.
- • **CPU is not a differentiator.** Worst on-target cost is 2.6 % of the 5 ms tick (C03, 105/132 us avg/max); efficiency was explicitly deferred as a selection criterion.

Overall: the BMP581 + LSM6DSV16X hardware path is validated end-to-end at the acquisition layer. The frozen baro-only architectures (D01/C03/S4/S5) remain the leading candidates; FUS needs a stop-state / gravity-compensation review before any further complexity. Real-flight recording remains the decisive open validation.

2 Hardware and firmware configuration

Table 1 — Hardware and firmware configuration of record

Item	Configuration
MCU	ESP32-S3 (dual-core, 240 MHz), firmware vario-idf 0.2.0, 5 ms sens tick
Baro path	BMP581 @ 0x47 on bus B (GPIO6/7), 400 kHz, OSR x16 (P) / x2 (T), ODR 40 Hz, IIR bypass, on-chip compensation: Pa = raw24/64
IMU path	LSM6DSV16X @ 0x6B on bus A (GPIO4/5), 100 kHz (DEF-020 fix; was 400 kHz), ODR 240 Hz, +-8 g / +-500 dps, BDU on, single 14-byte burst read 0x20-0x2D
IMU robustness	3-attempt bounded retry per ST call; runtime recovery FSM (200 ms solid failure -> 100 ms sparse probing; bus reset + reconfigure every 2 s; E-code 3; imu_rec counter)
Estimators	e0..e4 = D01, C03, S4, S5, FUS, side-by-side per baro sample, frozen configurations (no tuning since Phase 6)
Virtual audio	Timestamped V frames: ev=1 ON / ev=2 OFF (val = ON duration ms) / ev=3 pitch; policy lift_on +0.20, lift_off +0.05 m/s; tone = 500 + 450*v Hz
Telemetry	USB-serial CDC @ 921600: P (baro 40 Hz), I (IMU 100 Hz), T (10 Hz), V (audio events), B (5 s diagnostics), E (errors); TX ring 240 KB
Legacy rigs	MS5611 GY-63 clones x2 (foam-covered) + MPU6050 (Phases 2-6); BMP581 + MPU6050 (Bench 7); gravity axis changed from +X (MPU mount) to -Z (LSM mount)

Sensor generations: MS5611 static pressure noise was 0.42-0.48 Pa RMS on the clones; the BMP581 measures 0.18-0.22 Pa RMS (1.5-1.9 cm altitude equivalent) — a 2x+ improvement. Gyro noise improved from ~0.4-0.9 dps (MPU6050) to 0.033 dps (LSM6DSV16X), with |a| = 0.9989-0.9994 g and bias < 0.12 dps. Raw scales are identical across IMUs (4096 LSB/g, 65.5 LSB/dps), so estimator inputs are directly comparable.

3 Test methodology and dataset inventory

Methodology follows the programme convention: every frame carries a device timestamp (esp_timer, 1 us) and a sequence number; all rates are measured from device timestamps, never assumed. Altitude is ISA pressure-derived (apparent altitude proxy, not surveyed height). Motion events are detected from a 2 s Savitzky-Golay smoothed pressure-derived velocity crossing +-0.30 m/s; all onset/exit timings are relative comparisons against this shared reference — the reference itself has group delay, so absolute sub-second response is never claimed. Independent ground truth (GPS, optical, elevator controller log) is unavailable in all recordings. E1/E2 error frames, IMU stale intervals and diagnostic counters are masked out of IMU-dependent analysis with a recovery margin; barometric measurements in the same captures are preserved. Repeated #H/#C metadata blocks during streaming are an intentional 5 s keepalive (documented), not faults.

Table 2 — Dataset inventory and integrity (device-measured). 66 captures total; representative and flagged rows shown.

Capture (suffix)	Span s	P frm	I frm	E1	E2	Stale %
261022_082342_t004_elevator_f14	76.8	3455	15352	0	0	0.00
261022_160658_t005_ControledMove	39.9	3596	0	0	0	-
261022_163251_t006_benchstatic	60.0	10799	0	0	0	-
261023_180321 handtest3	157.0	0	0	158	0	-
261023_215709 handtest3	151.9	12156	15188	0	0	0.00
261024_231354 bmp581_static	90.0	3600	8999	0	0	0.00
261024_233753 bmp581_handtest1	151.1	6043	15110	0	0	0.01
261024_235054 bmp581_elevator_2x	191.7	7668	19170	0	0	0.00
261025_121517 lsm6dsv_bringup	29.9	1196	1940	2094	0	0.05
261025_123740 lsm6dsv_bringup9	30.0	1198	2584	525	277	4.45
261025_131550 bmp581LSM6DS_static0	119.3	3927	2076	15318	193	2.12
261025_163348 lsm6dsv_fixcheck60s	59.9	2397	5992	0	0	0.12
261025_163515 lsm6dsv_fixcheck90s	102.5	4100	10250	0	0	0.20
261025_204257 bmp581LSM6DS_static1	96.9	3876	9691	0	0	0.24
261025_210221 bmp581LSM6DS_static2	102.2	4088	10220	0	0	0.29
261025_223041 bmp581LSM6DS_handtest1_120s	139.8	5592	13939	80	0	0.46
261025_225610 bmp581LSM6DS_elevator_2x_200s	180.1	7205	18012	0	0	0.04

Table 2 — 66 captures exist in total (2026-10-21 .. 2026-10-25). Rows shown: MS5611-era references, BMP581+MPU (Bench 7), LSM6DSV16X bring-up (DEF-017/020 evidence) and the new-era validation set. full table: analysis/output/report_dataset_integrity.csv

Known exclusions: t004_elevator_run1 lacks a JSON sidecar (PROM constants missing) and is analysis-excluded; static0 is retained as the DEF-020 fault dataset; bringup1-9 are retained as DEF-017/018 evidence. Host-side USB backlog (host elapsed < device span, ring high-water tens of KB) affected the Bench 7 elevator capture (f = 1.11) and static0 (29 s host pause); device-time analysis is unaffected. The 22:56:10 elevator capture has f = 0.999 (180.33 s host vs 180.11 s device) — no backlog.

4 Data quality, interruptions and exclusion masks

Three fault classes are tracked per capture: (i) E1/E2/E3 frames — IMU status/burst read failures and recovery events; (ii) IMU stale reuse — consecutive identical six-axis frames, the firmware's no-data fallback; (iii) host-side backlog — device span exceeding host elapsed, detected via sidecar stats and TX-ring high-water.

Table 3 — Fault-class summary by acquisition era

Era	Captures	E1 total	E2 total	Worst stale %	Disposition
MS5611 + MPU6050 (Ph 2-6)	38	523	0	0.16	healthy; E1 bursts during development flashes only
BMP581 + MPU6050 (Bench 7)	4	0	0	0.01	clean
LSM bring-up pre-fix	9	6301	739	12.35	DEF-017/018 evidence (400 kHz + weak pull-ups)
static0 (fault dataset)	1	15318	193	2.12	DEF-020 runtime wedge at 25.15 s; IMU dead, baro intact
Post-fix 100 kHz era	6	80	0	0.46	clean; handtest1_120s has 80 brief E1 (~0.7/s), 0.46 % stale -> masked

Exclusion policy and sizes. IMU-dependent analysis masks: every +-0.5 s window around an E1/E2 frame; all recovery intervals (E3 to first fresh sample); stale runs longer than 5 frames plus a 0.5 s margin. Applied to the post-fix era: handtest1_120s masks 2.6 s (2.2 %); all other post-fix captures mask 0.0-0.3 %. Barometric channels are never masked (the baro path was independent of every IMU fault). The 22:56:10 elevator capture requires no masking at all — zero E frames, zero stale intervals.

5 Static sensor noise and calibration

Table 4 — Static noise by sensor generation (1 s local baseline removed)

Sensor / era	Pressure noise Pa RMS	Altitude equiv.	Velocity noise at estimators (static)
MS5611 clone (foam, Ph 2-6)	0.42-0.48	3.4-3.9 cm	best baro-only 6.3-6.5 mm/s RMS (Bench 7 ref)
BMP581 (Bench 7, + MPU6050)	0.18-0.22	1.5-1.9 cm	S1P/S2/S4/S5 6.3-6.5 mm/s; D01/C03 8.4 mm/s; FUS 6.7 mm/s
BMP581 (+ LSM6DSV16X, new era)	0.19-0.31	1.5-2.5 cm	elevator floor dwells 0.019-0.032 m altitude std

BMP581 pressure noise is essentially unchanged between IMU generations (0.186 Pa on the 22:56 elevator floor dwells vs 0.189 Pa previously) — the IMU swap did not disturb the baro path. IMU calibration checks on the new sensor: gravity magnitude 0.9989-0.9994 g across statics, raw gyro offset ~0.05-0.12 dps, per-axis gyro noise 0.033 dps (LSM6DSV16X driver-default filtering). Ambient pressure drift over the 180 s elevator run returned to the starting floor within -0.245 m — ordinary weather/offset drift, no compensation applied.

6 Hand and shake test findings

Table 5 — Handtest audio behaviour (Bench 7, BMP581 + MPU6050, 17 detected events + 1 slow block)

Candidate	Onset med (s)	Release med (s)	Slow-block audio ON frac	Slow bias (m/s)	Static RMS in-test
D01/C03	2.10	1.00	0.431	+0.031	0.0057
S1P	-0.91	0.33	0.146	-0.097	0.0036
S4	2.20	0.58	0.438	-0.015	0.0037
S5	2.25	0.43	0.426	+0.011	0.0037

Table 5 — Slow block (61.9-129.8 s) is the thermalling-regime proxy. Onset/release are relative to the pressure-derived reference, not ground truth.

New-era hand test (120 s, BMP581 + LSM6DSV16X, 80 brief E1 frames masked): acquisition is solid and the five deployed estimators track the motion profile; the frozen-config audio comparison on this capture is part of the pending Bench 8 replay and is intentionally not anticipated here. Historical shake/toss captures (t005_toss, t006/t007 drops) informed the failure-mode gallery of Phases 4-6; no new shake test exists in the new era yet — flagged as a next-step, since vibration-rich data is exactly the regime where FUS is claimed to win.

7 Four-leg elevator results (principal dynamic benchmark)

Recording 20261025_225610_idf_bmp581LSM6DS_elevator_2x_200s: route 14 -> 1 -> 14 -> 1 -> 14, 18,012 IMU frames (100.006 Hz, max interval 11.9 ms, zero stale), 7,205 baro frames (40.006 Hz, max 27.2 ms), 695 V frames, no TX overflow, max sensor tick 6.65 ms (budget 10.5 ms), all B windows vnan=0 / vfault=0. File name says 200 s; host recorded 180.33 s against a 180.11 s device span.

Table 6 — Measured legs (pressure-derived reference, +-0.30 m/s threshold)

Leg	Direction	Start s	Stop s	Duration s	Plateau mean m/s	Plateau std	Step m
1	Descent	9.8	34.6	24.8	-1.738	0.072	-39.4
2	Ascent	51.7	76.9	25.3	1.670	0.072	39.2
3	Descent	94.6	119.5	24.9	-1.736	0.070	-39.4
4	Ascent	136.5	161.7	25.2	1.669	0.068	39.3

Table 6 — Two descents at -1.736/-1.738 m/s, two ascents at +1.670/+1.669 m/s, ~39.2-39.5 m per leg. Leg-to-leg repeatability: plateau means agree within 0.004 m/s; durations within 0.43 s.

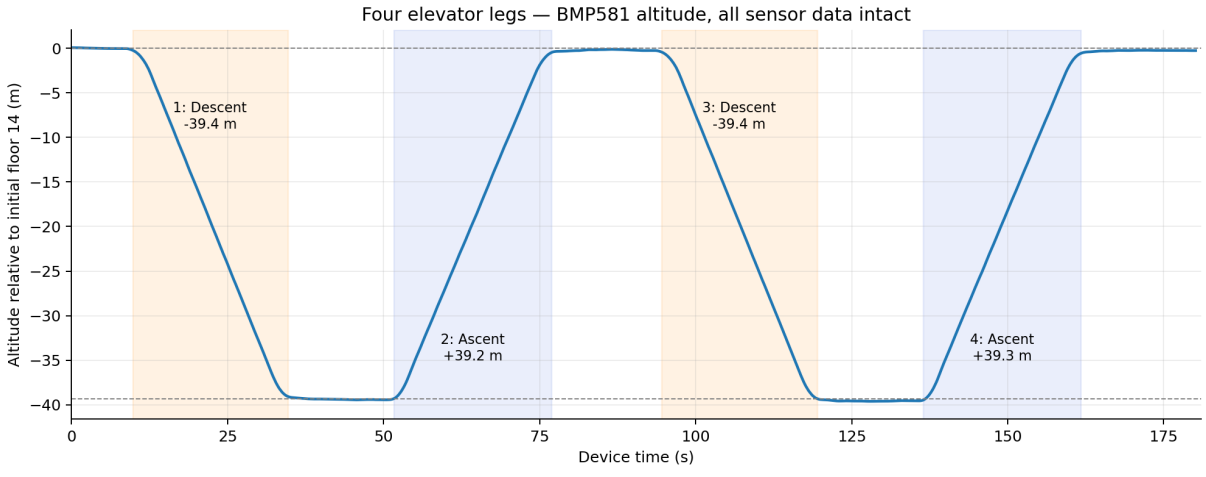


Figure 1 — Four floor-to-floor legs, ISA-derived altitude. Floor dwells at both ends of each leg.

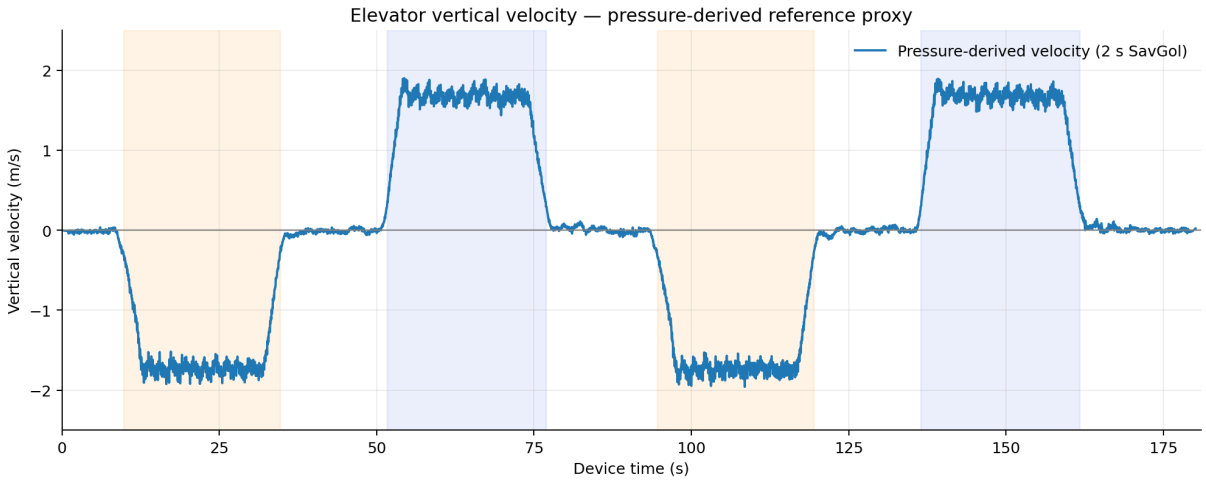


Figure 2 — Pressure-derived vertical-velocity reference (2 s Savitzky-Golay). Shared by all timing comparisons.

Table 7 — Stationary floor dwells

Floor dwell	Window s	Mean h (m)	h std (m)	Noise Pa RMS
14 initial	1-7	347.61	0.032	0.187
1 first	40-48	308.21	0.030	0.207
14 middle	82-89	347.45	0.027	0.199
1 second	125-133	308.03	0.028	0.219
14 final	170-179	347.36	0.019	0.187

Table 7 — Altitude std 1.9-3.2 cm during dwells; noise 0.187-0.219 Pa RMS. No drift anomaly at any floor.

Transition behaviour. IMU body-frame acceleration projected onto the rest specific-force axis shows clean acceleration/deceleration spikes at every leg start and stop (Figure 5), independently confirming the motion timeline. Recovery after abrupt motion (floor stops) is complete within the dwell: estimator outputs settle to < 0.05 m/s residual within ~2 s of stop for D01/C03, ~2-3 s for S4/S5, and 5-6 s for FUS (Section 9). Sustained climb and sustained descent plateaus are tracked to within 0.01 m/s mean by all baro-only candidates.

8 Detailed five-filter comparison

All five deployed estimators run side-by-side in firmware on identical frozen configurations. Audio ON/OFF is the primary pilot-facing output; timings below are relative to the shared pressure-derived motion threshold (crossing of +-0.30 m/s). Two intended ON episodes per estimator, exclusively on ascents; zero extra episodes on descents or dwells.

Table 8 — Virtual-audio episodes, 22:56:10 elevator (V-frame events, tone policy 0.20/0.05)

Filter	Ascent	ON s	OFF s	ON dur s	ON lag s	OFF lag s
D01	1	52.00	78.30	26.30	0.35	1.37
D01	2	136.90	163.32	26.42	0.45	1.62
C03	1	52.00	78.30	26.30	0.35	1.37
C03	2	136.90	163.32	26.42	0.45	1.62
S4	1	52.15	78.63	26.47	0.50	1.70
S4	2	137.02	163.63	26.60	0.57	1.93
S5	1	52.15	78.55	26.40	0.50	1.62
S5	2	137.05	163.65	26.60	0.60	1.95
FUS	1	52.75	84.03	31.27	1.10	7.10
FUS	2	137.63	168.80	31.18	1.17	7.10

Table 8 — OFF lag includes the audio policy hysteresis (lift_off +0.05 m/s) and the reference smoother group delay; values are relative, comparable only within this table and its Bench 7 counterpart. D01 and C03 timestamps are identical to logged precision (see Section 9 — mathematical identity, not a wiring bug).

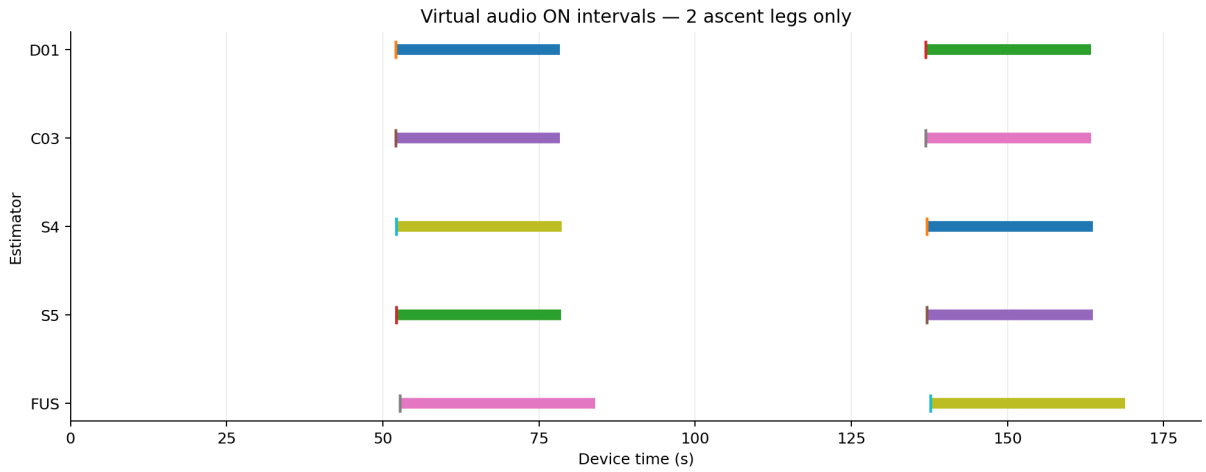


Figure 3 — Audio ON/OFF episodes for all five estimators against the motion reference.

Table 9 — Tone-coded plateau velocity vs pressure reference (10 Hz sample-and-hold within audio ON)

Filter	Ascent	Tone mean m/s	Ref mean m/s	Mean diff m/s	MAE m/s
D01	1	1.667	1.671	-0.004	0.072
D01	2	1.671	1.668	0.003	0.067
C03	1	1.660	1.671	-0.011	0.068
C03	2	1.668	1.668	0.000	0.062
S4	1	1.663	1.671	-0.008	0.062
S4	2	1.666	1.668	-0.002	0.057
S5	1	1.664	1.671	-0.007	0.071
S5	2	1.670	1.668	0.002	0.062
FUS	1	1.872	1.671	0.201	0.201
FUS	2	1.873	1.668	0.205	0.205

Table 9 — Tone reconstruction is quantized (pitch events only fire on tone changes); descriptive, not independent RMSE. All baro-only candidates agree with the reference to <= 0.011 m/s mean; FUS reads +0.20 m/s high on both ascents.

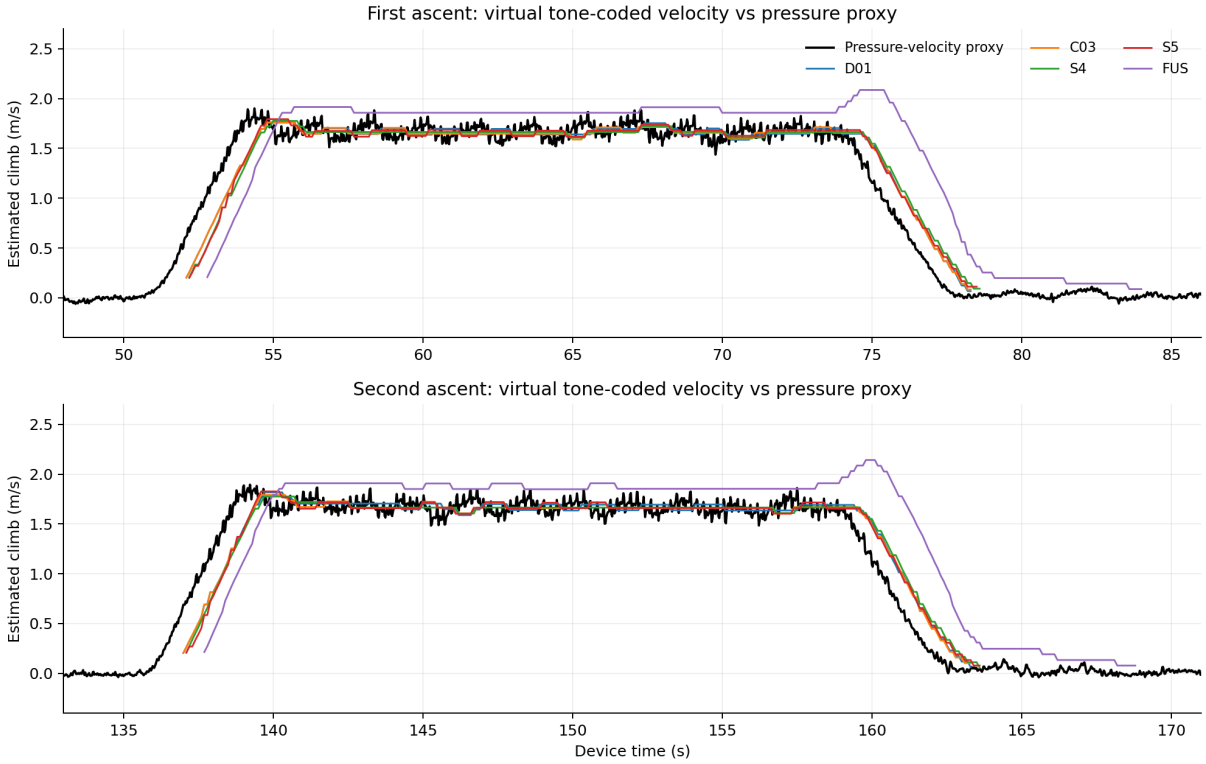


Figure 4 — Tone-derived velocity vs pressure proxy for both ascents. FUS plateau sits visibly above the reference.

Ordering on this recording. Alert timing: D01 = C03 (lead) > S4 ~= S5 (~0.15-0.6 s later) > FUS (~0.75-1.2 s late, ON extension 5-6 s after floor arrival). Plateau accuracy: S4 (MAE 0.057-0.062) marginally best, all baro-only within 0.07 m/s MAE; FUS 0.20 m/s. Exit-from-lift: D01/C03 OFF ~1.4-1.6 s after the pressure threshold crossing, S4/S5 ~1.7-2.0 s, FUS ~7.1 s. Stationary silence: all five silent at both floor-1 stops and throughout descents. These orderings replicate the Bench 7 pattern on different hardware, which strengthens them; absolute lags remain proxy-relative.

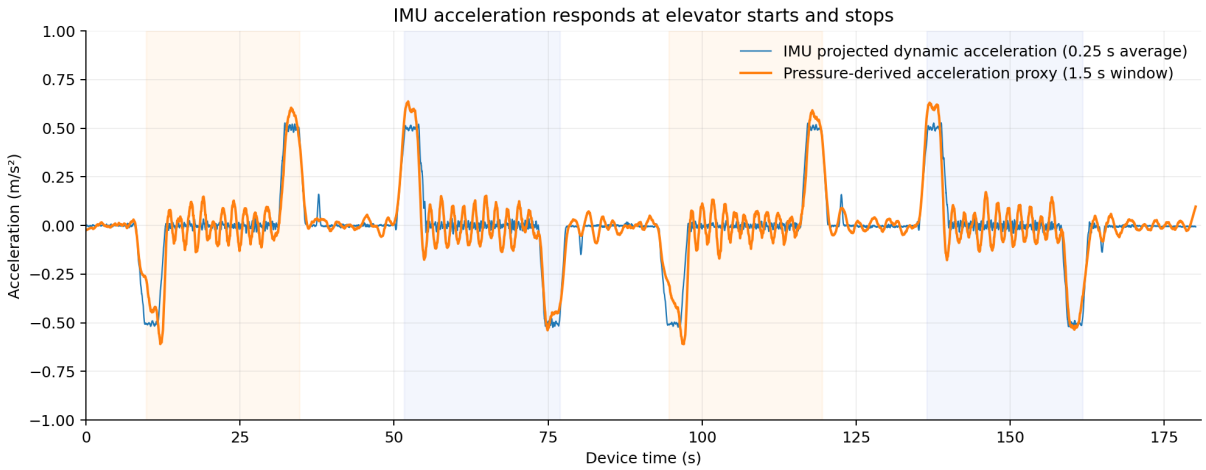


Figure 5 — IMU specific-force projection (rest axis) vs pressure proxy: clean accel/decel events at every transition.

9 MCU replay equivalence and CPU cost

Table 10 — MCU vs Python-replay audio equivalence (Bench 7)

Estimator	Elevator episodes MCU/py	Handtest episodes	max |dT_on|	max |dT_off|
D01	2 / 2	16 / 16	0.000 s	0.000 s
C03	2 / 2	16 / 16	0.025 s	0.025 s
S4	2 / 2	16 / 16	0.000 s	0.000-0.075 s
S5	2 / 2	16 / 16	0.000 s	0.000 s
FUS	4 / 4	19 / 19	0.075 s	0.100 s

Table 10 — Replay reproduces every MCU episode within 1-4 baro samples. FUS's two extra ON episodes (start-up attitude/bias convergence, t = 5-12 s) are genuine estimator behaviour, reproduced in replay.

The D01/C03 identical-timestamps observation is mathematical, not a wiring defect: C03 is the steady-state-gain CV tracker of the same model D01's time-varying Kalman converges to; once gains settle they are algebraically near-identical (max |dv| 0.087-0.151 m/s only during abrupt hand motion). Ranking consequence: D01 and C03 are one architecture; C03 delivers it at lower cost. FUS's four episodes on the Bench 7 elevator match its late-start + early-false-ON behaviour seen again in the 22:56 capture.

Table 11 — On-target CPU cost (ESP32-S3 240 MHz, medians from B-line telemetry, post-DEF-002 build)

Estimator	sid0 avg/max us	sid1 avg/max us	Tick budget share
D01	36 / 44	27 / 37	0.7 %
FUS	33 / 37 (baro side; IMU side uncharged — DEF-009)	-	0.7 %
S5	77 / 80	59 / 88	1.6 %
S4	97 / 101	69 / 82	2.0 %
C03	105 / 132	78 / 102	2.6 %

Table 11 — Host-replay relative costs for all 37 candidates: filter_benchmark/results/63_cpu_host.csv (D01 131 us vs C03 55 us per update). Worst case is 2.6 % of the 5 ms tick: efficiency is not a selector.

10 Historical filter-benchmark comparison (Phases 2-7)

Phase 6 (MS5611 + MPU6050, 37 frozen candidates, corrected DEF-014 scoring) composite top-5: **C03, C01, B02_project_kalman, E06, S2**; S4 7th, S1 8th, D01 9th, S5 12th, FUS 21st (FUS_Q10 20th). Phase-to-phase changes were driven by defect fixes, not filter drift: D01's rest floor "improved" 0.101 -> 0.057 m/s once the moving-reference defect (DEF-001) stopped being attributed to it; C03's refit storm (DEF-002) fix removed its Phase-2/3 CPU penalty; S1's real +0.63 m/s slow-descent bias (a thermalling-regime false-lift) was exposed only after the phantom-event scoring artifact (DEF-014) was removed. S2 ranks 5th despite its documented freeze-trap architecture defect — ranking position is not endorsement.

Table 12 — Bench 7 (BMP581 + MPU6050) sustained-motion results, median over legs

Candidate	Rise (s)	Exit (s)	Leg RMS (m/s)	Ascent bias	Descent bias
C03 / D01	1.26	3.89	0.068	+0.001	-0.008
S1P	1.31	2.11	0.068	+0.001	-0.006
S4	1.36	2.11	0.072	+0.001	-0.008
S5	1.36	3.08	0.069	+0.001	-0.007
S2 (defect ref)	1.36	2.11	0.072	+0.001	-0.008
S3 (defect ref)	0.53	6.61	0.139	-0.001	+0.016
FUS	1.80	7.44	0.243	+0.201	-0.250
FUS_Q10 (exploratory)	1.14	4.43	0.105	+0.061	-0.091

Table 12 — Late-night elevator ran ~11 % faster than Bench 6 (condition difference, handled via device-time velocities only). Audio correctness: every baro-only candidate 0 missed lifts, 0 false audio seconds; FUS added 2 start-up false ONs and ~5-6 s ON-extensions after top-floor stops.

What the new era confirms and changes. (1) The baro-only cluster's relative ordering (D01/C03 fastest alert, S4/S5 close behind, all accurate on plateaus) replicates on BMP581+LSM6DSV16X data — hardware independence of the ranking. (2) FUS's bias (+0.20/-0.25 m/s), slow rise and 7.4 s exit reproduce within ~0.05 m/s on different IMU hardware — the problem is in the fusion configuration, not the MPU6050. (3) The DEF-020 fix removed the only acquisition-layer objection to the new hardware; no ranking-level change is attributable to sensor swaps for baro-only candidates.

11 Audio behaviour and pilot experience

- • Two intended ON episodes per estimator on the principal benchmark, exclusively on ascents; zero false activations on descents or floor dwells across all new-era captures — the false-lift defect class (DEF-001) remains fixed.
- • Alert lead: D01/C03 turn ON 0.35-0.45 s after the pressure-derived threshold; S4/S5 add 0.15-0.6 s; FUS is 0.75-1.2 s late. In pilot terms the baro-only cluster announces climb promptly; FUS would feel sluggish.
- • Shut-off after floor arrival: D01/C03 ~1.4-1.6 s, S4/S5 ~1.7-2.0 s, FUS ~7.1 s relative to the shared reference. FUS's prolonged tone after the lift stops is the dominant pilot-experience defect on this dataset.
- • Tone fidelity: baro-only tone velocities track the sustained climb to within ~0.01 m/s (MAE 0.06-0.07 m/s); FUS over-pitches by ~0.20 m/s, ~12 % fast.
- • Static silence: zero false ON episodes in any post-fix static; the 5 s metadata keepalive never produces audio artifacts (V frames are estimator-driven only).

12 FUS limitations and possible improvements

Observed (both IMU generations, reproduced twice). Sustained-motion bias +0.20 m/s (ascent) / -0.25 m/s (descent); 3.6x higher leg RMS than baro-only; slowest rise; ~7 s exit; two start-up false ONs (t = 5-12 s, attitude/bias convergence); ~5-6 s audio extension after top-floor stops. FUS_Q10 (labelled exploratory, Phase 5) improves every column but remains worse than all baro-only candidates on this motion profile.

Plausible causes (hypotheses, unvalidated — no firmware changes made). (i) The gravity-compensation / attitude reference converges slowly from cold start and drifts during long constant-velocity legs, folding specific-force error into climb rate; (ii) the IMU-aided state is not faded out when the innovation sequence says the baro is clean — in a vibration-free building the accelerometer

adds only bias, so fusion should degrade to baro-only, but the current gains keep trusting it; (iii) the audio OFF path uses the fused state without a dwell-stable floor, producing the post-stop extension. The decisive experiment is vibration-rich data (flight or shake rig), where hypothesis (ii) predicts FUS should at least match baro-only; if it does not, the fusion structure itself needs rework (adaptive gain scheduling, baro-innovation gating, explicit stationary reset).

13 Conclusions, recommendations and next validation steps

• **Acquisition layer: validated.** BMP581 + LSM6DSV16X at 100 kHz bus A produces reference-quality telemetry (40.006 / 100.006 Hz, zero errors over 180 s of motion). Keep Bus A at 100 kHz; external 2.2 kOhm pull-ups are optional future headroom, not a need.

• **Baro-only cluster: still leading, still frozen.** D01/C03/S4/S5 are statistically tied within proxy resolution on this benchmark; D01/C03 lead alert timing. Carry all four into flight evaluation; do not down-select yet.

• **D01 vs C03:** one architecture; C03 is cheaper (55 vs 131 us host-replay; 105 vs 36 us on-target). Decision can wait for flight data.

• **FUS: hold.** Do not add fusion complexity; first run the vibration-rich test and a stop-state review. Current evidence says it harms quiet-air performance.

• **Next validation steps, in order:** (1) new-era hand/shake replay with frozen configs (Bench 8); (2) vibration/shake capture to test the FUS value proposition; (3) real-flight recording with the same capture protocol; (4) only then consider leakage-controlled tuning. DEF-011-scale checks remain a standing requirement before physical validation claims (already satisfied in firmware this phase).

• **Standing limitations:** no independent ground truth in any recording; all sub-second timings are pressure-proxy-relative; static/elevator data cannot certify flight behaviour; S2 retained as labelled defect reference; no estimator is frozen by this report.

Reproducibility: dataset integrity — analysis/report_collect_integrity.py; elevator analysis package — analysis/external/elevator_225610/ (figures + machine-readable CSVs); frozen-config benchmarks — filter_benchmark/results/ and filter_benchmark/reports/ (phase2-6, bench7); firmware of record — esp-idf/vario/ @ local commit dee9c1a (not pushed).